



Experiential Study Tour Murchison Falls National Park.



Experiential Study Tour Murchison Falls National Park.

A Journey of Learning and Discovery 2025

Kibwota Welborn

August 8, 2026



Background of the Study

Murchison Falls National Park, located in North-western Uganda, is both historically and ecologically significant. It was gazetted in 1952, making it Uganda's oldest and largest protected area. The park covers an estimated 3,893 km², spanning across the districts of Nwoya, Buliisa, Kiryandongo, and Masindi, and is bisected by the River Nile.

The falls themselves represent one of the most iconic natural wonders in East Africa. The Nile, the world's longest river, is forced through a narrow 7-meter gorge, dropping with immense power and creating a thunderous roar before flowing into Lake Albert. This hydrological process not only provides an inspiring tourist attraction but also symbolizes the resilience and force of natural systems. Historically, the park has hosted early explorers such as Sir Samuel Baker in the mid-19th century, who named the falls after Roderick Murchison, then-president of the Royal Geographical Society. Over time, Murchison Falls has become a focal point for wildlife conservation, eco-tourism, and academic research, especially because of its rich biodiversity that includes 76 mammal species and over 450 bird species.

Field trips like the one conducted in September 2025 are designed to immerse learners, researchers, and tourists in a practical environment where theoretical knowledge can be tested against real ecological and socio-economic realities. They offer

opportunities to observe ecosystems in their natural states, interact with conservation staff and local communities, and understand the challenges of balancing tourism with environmental conservation.



*The two falls are seen from the angle
716mEyeAlt,Elev 632m (2°16'33.64"N, 31°41'0.31"E)*

Literature Review of Murchison Falls National Park

Historically, the park gained prominence during the 20th century as a safari destination. It is home to more than 76 mammal species

(including the “Big Five”) and over 450 bird species and one of the oldest nation park in Uganda.([Uganda Wildlife Authority](#).) The falls are the park’s defining feature, where the Nile’s force demonstrates both aesthetic beauty and hydrological significance.Murchison Falls National Park's history dates back to 1907 when the area was cleared due to sleeping sickness and later became a game reserve in 1910. In 1952, the park was officially established as a national park under the ([Uganda National Parks Act](#)), making it one of Uganda's oldest conservation areas. Notable explorers like ([John Speke](#), [James Grant](#)), and ([Samuel Baker](#)) visited the area in the 1860s, with Baker naming the falls Murchison Falls after the geologist ([Roderick Murchison](#)).The establishment and conservation of Murchison Falls National park started by massive evacuation due to tsetse flies infestations which was causing sleeping sickness in a spanning area of approximately 13,000 km² making it inhabitant for settlement and farming in 1907-1912.

Significance of the Study

This field research contributes to academic knowledge, practical learning, and environmental awareness. It offers insights into conservation, eco-tourism, and community development while serving as a reference material for future research and policy discussions.

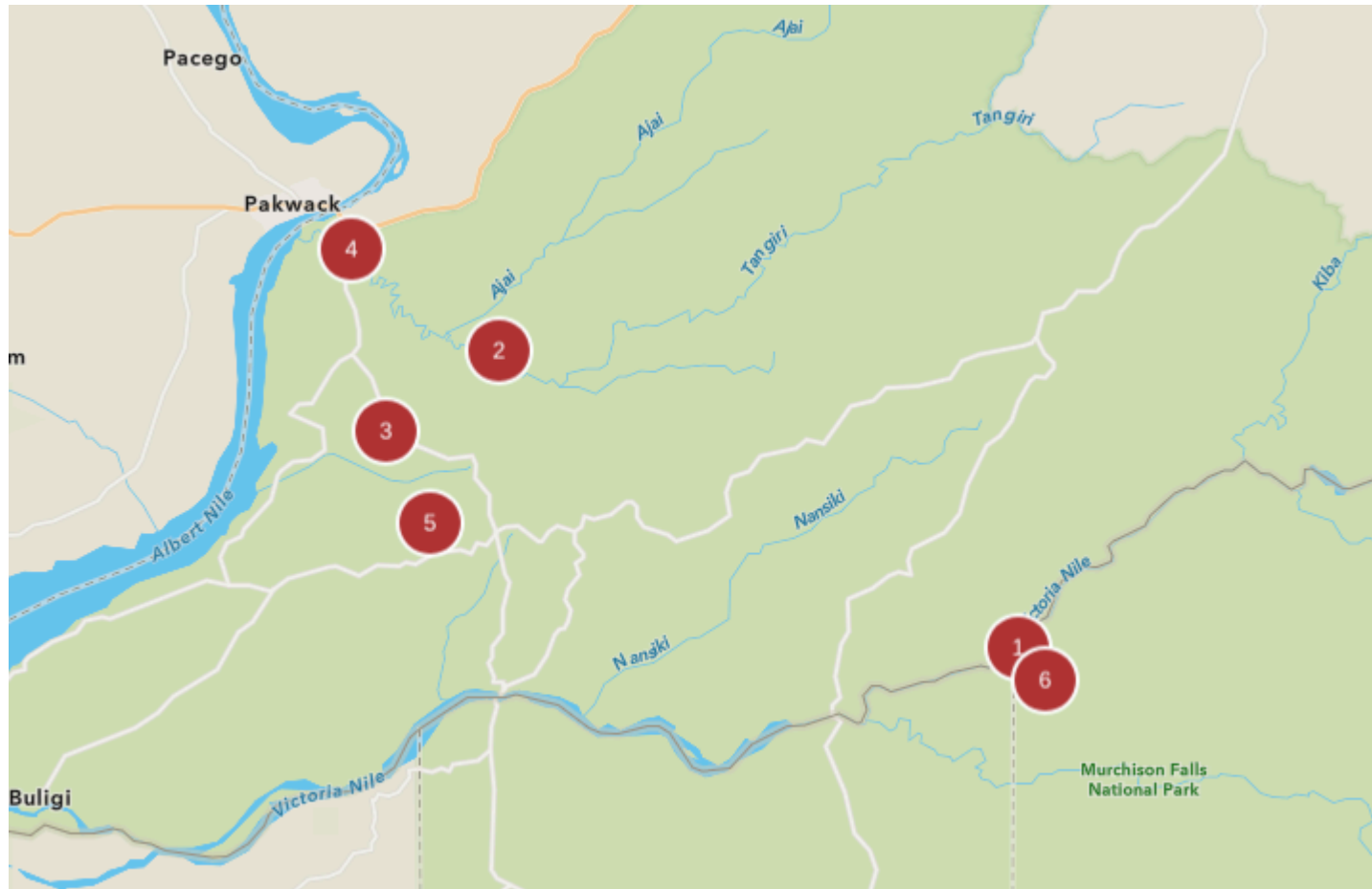
The significance of this study extends beyond the immediate experience of participants. It lies in its contribution to knowledge production, environmental awareness, socio-economic development, cultural understanding, and policy formulation. The trip to Murchison Falls National Park serves as a model for how experiential learning can be used not only to strengthen education but also to contribute to broader discourses in conservation, tourism, and sustainable development.

The significance of this study extends beyond the immediate experience of participants. It lies in its contribution to knowledge production, environmental awareness, socio-economic development, cultural understanding, and policy formulation. The trip to Murchison Falls National Park serves as a model for how experiential learning can be used not only to strengthen education but also to contribute to broader discourses in conservation, tourism, and sustainable development.

Academic and Educational Significance

Bridging Theory and Practice: The trip allowed learners to move beyond classroom-based knowledge to direct engagement with ecosystems. Concepts such as biodiversity, eco-tourism, food chains, and conservation policies, often abstract in lectures, were observed in real-world settings, we move and experienced a lot of physical in-touch with the soils,trees grasses and seeing of different bio-diversities especially when we were at the top of the

fall ,we managed to observe the different tree species around
,birds, and animals like Monkey,Elephants ,Hippopotamus,Uganda
Kob,Bufallos and many others.

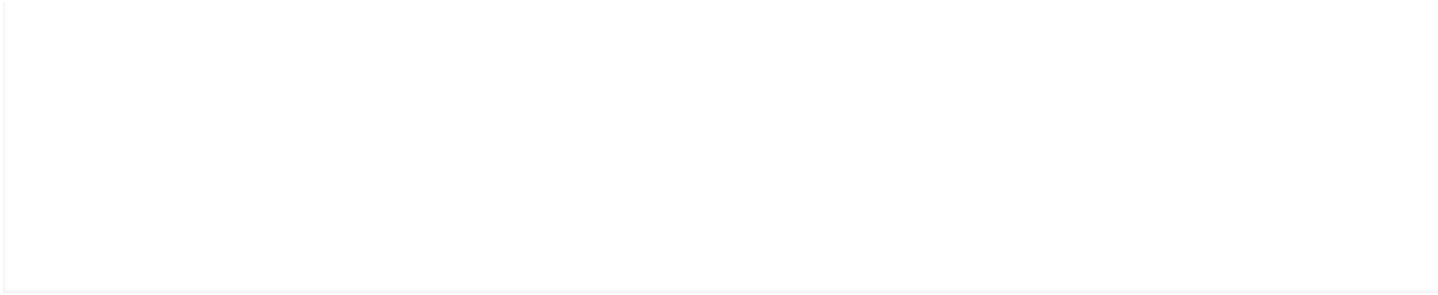


Esri | TomTom | Garmin | METI/NASA | USGS

5 mi | Powered by Esri

1

Monkey



Seen while we were at the top of the falls

2

Hippos in Swamp



Seen towards the Albert Nile near Pakwach Bridge

3

Giraffe



Seen During Game drive towards Albert Lake



Hippos



Identified as We head along Pakwach Bridge

5

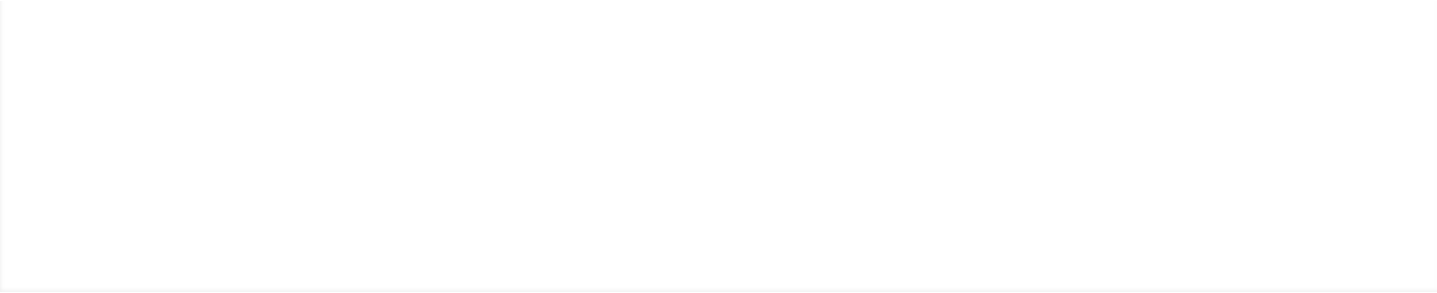
Elephant



Identified during game drive towards the Albert Lake



Uganda Kob

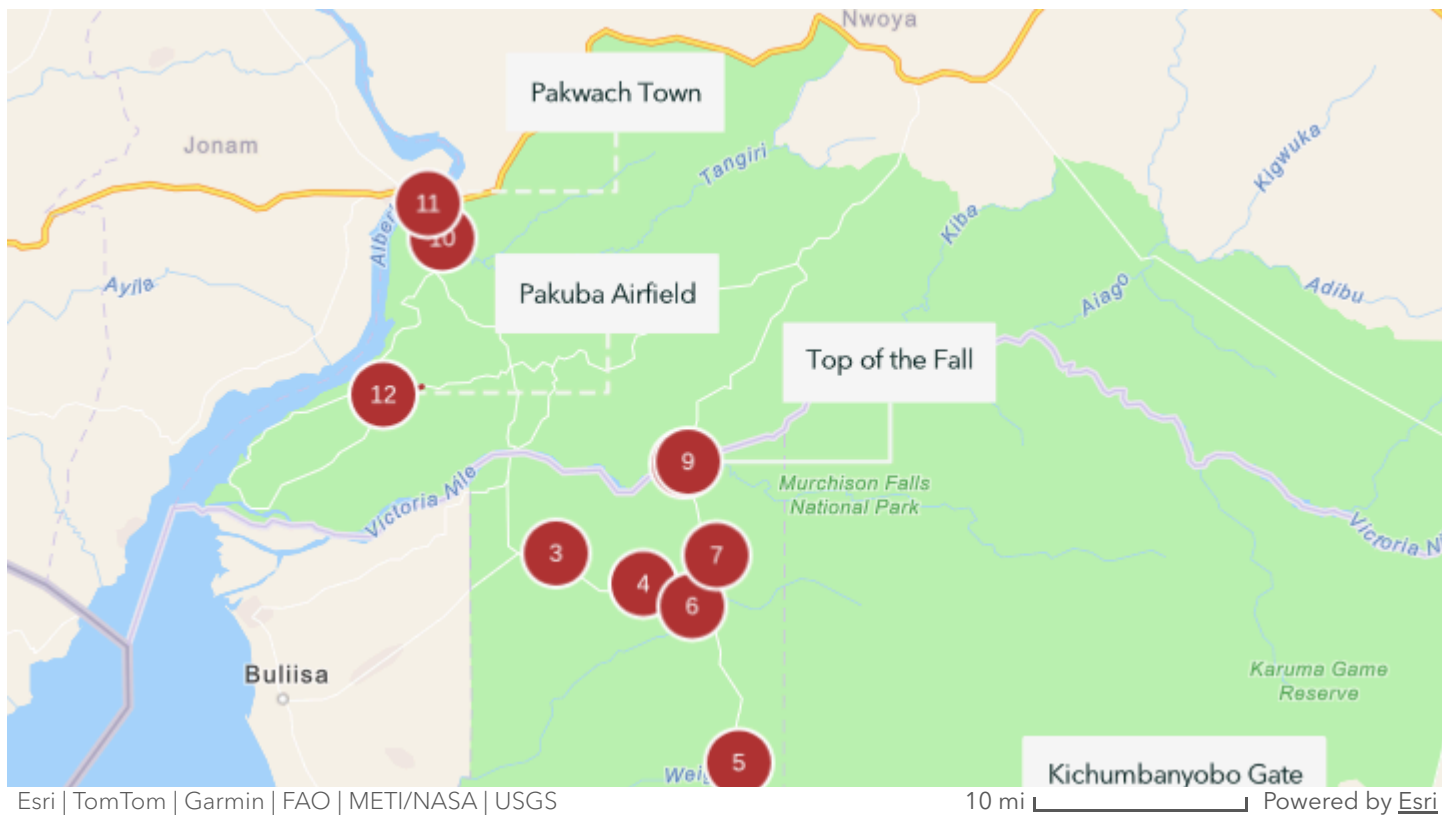


Seen in various places ,During game drive and Towards the Top of the falls in Sambiya Village

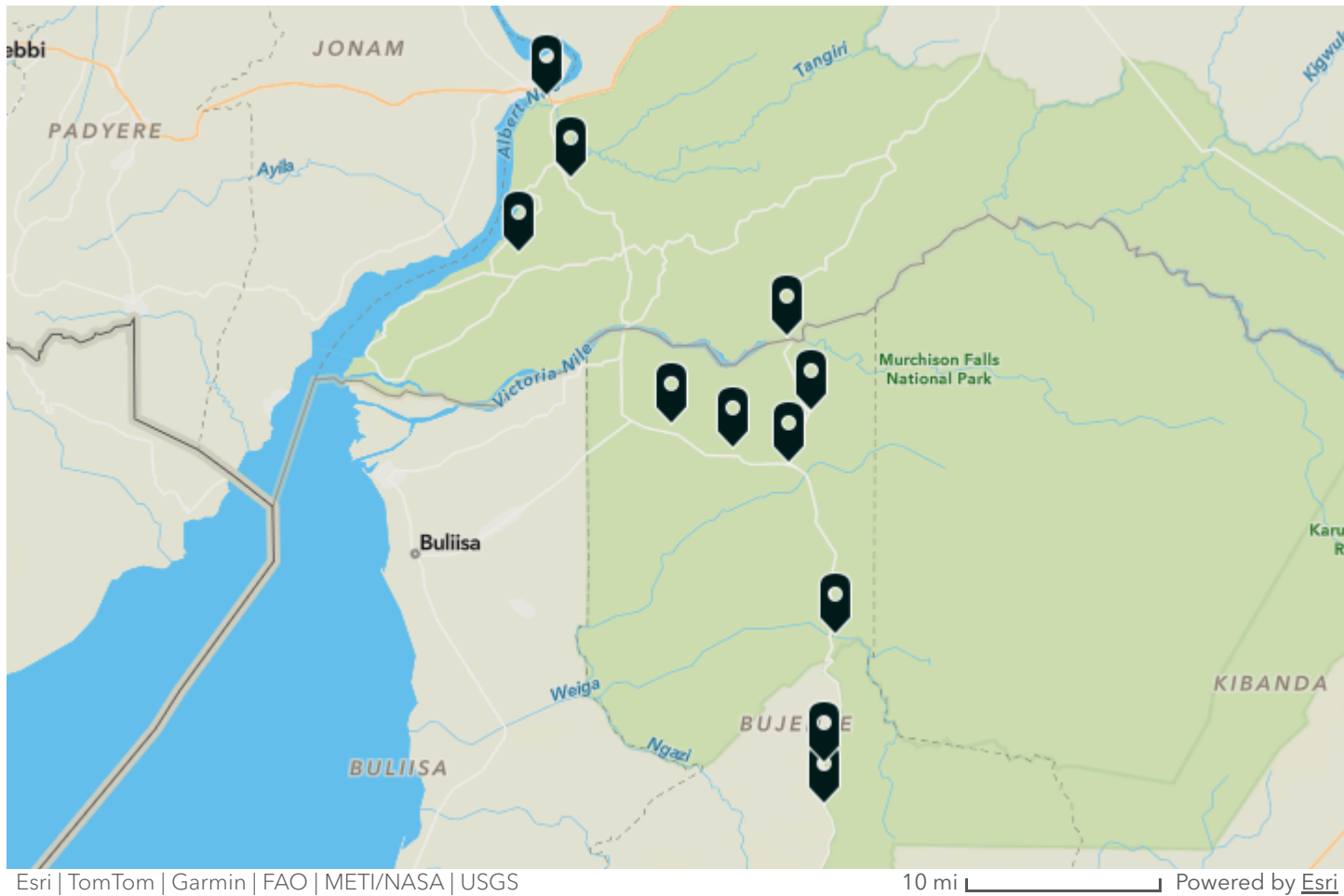
Experiential Learning: Guided by Kolb's Experiential Learning Theory, the trip fostered observation, reflection, and critical analysis. Students developed field research skills, such as species identification, data recording, and reflective journaling especially during the early morning game drive towards Pakuba Lodge ,Albert Nile , students were able to differentiate different animal species ,others were using Local Language as they look for English name and as well to identify the different vegetation zonation of the studied area as seen above and others seen below:

Hard wood vegetation types to soft wood vegetation types from the Entrance of Budongo Central Forest to Murchison Falls National Park 52.8KM to Top of the Fall.

Vegetation Zonation of Murchison Falls National Park



From Masindi Gate to the Top of the Falls and Towards Pakuba



Kichumbanyobo Gate / Budongo Forest edge

Driving from Masindi town to the Kichumbanyobo gate, you pass through Budongo forest tropical moist semi-deciduous forest, historically known for chimpanzee populations and...

Woodland transition zone

South of the main savanna belt, the terrain shifts to Combretum Terminalia woodland broad-leaved deciduous trees with a grass understorey, marking the ecotone between for...

Tall grass (Hyparrhenia) savanna

Rolling grasslands and acacia savannahs dominate the central stretch tall Hyparrhenia-type grasses interspersed with scattered acacia and other fire-tolerant trees, shaped by the...

Sambiya Junction

Rolling grasslands and acacia savannahs dominate the central stretch tall Hyparrhenia-type grasses interspersed with scattered acacia and other fire-tolerant trees, shaped by the...

Road cutting Towards the Top of the Fall

The dark, jagged exposed surface is weathered rock/shale substrate cut through during road construction this is consistent with the rocky, thin-soiled terrain that defines the...

Borassus palm savanna

A distinctive zone dominated by Borassus palms scattered through grassland one of the most photogenic and recognizable vegetation types in the park, often used as a signature...

Riverine forest / Nile corridor

Approaching the river, vegetation shifts to riverine forest and woodland fringing the Victoria Nile denser, greener growth sustained by proximity to water, forming part of the...

Seasonal Swamp Spotted with Buffalos

In the wet season water depth reaches 0.3–0.8 metres, while in the dry months (December–February) the outer edges recede, exposing bare ground useful if your field visit happens...

Borassus palm savanna

A distinctive zone dominated by Borassus palms scattered through grassland one of the most photogenic and recognizable vegetation types in the park, often used as a signature...

Floodplain zone (Towards Pakwach Bridge)

As the river widens beyond the falls toward the delta, vegetation shifts to wetland dominated by Cyperus papyrus, interspersed with patches of Vossia cuspidata and Ludwi...

Dry savanna near Pakuba

Pakuba sits on the eastern bank of the Albert Nile at the northwestern edge of the park, opening onto sweeping vistas of savannah plains. Vegetation here shifts back to open, dr...

The trip fostered refectory observation on the strength of the water falls

I myself observed two water falls ,one being stronger than the other, but both from the same flowing Nile river ,the first fall being identified as the "**Kabalega fall**" which is the ancient fall formed due to tectonic processes and it extends as deep as **45m**

inside the gorge, the water falls at a high speed and of large volume.

MURCHISON Falls Strong water flows of over 3000m/s

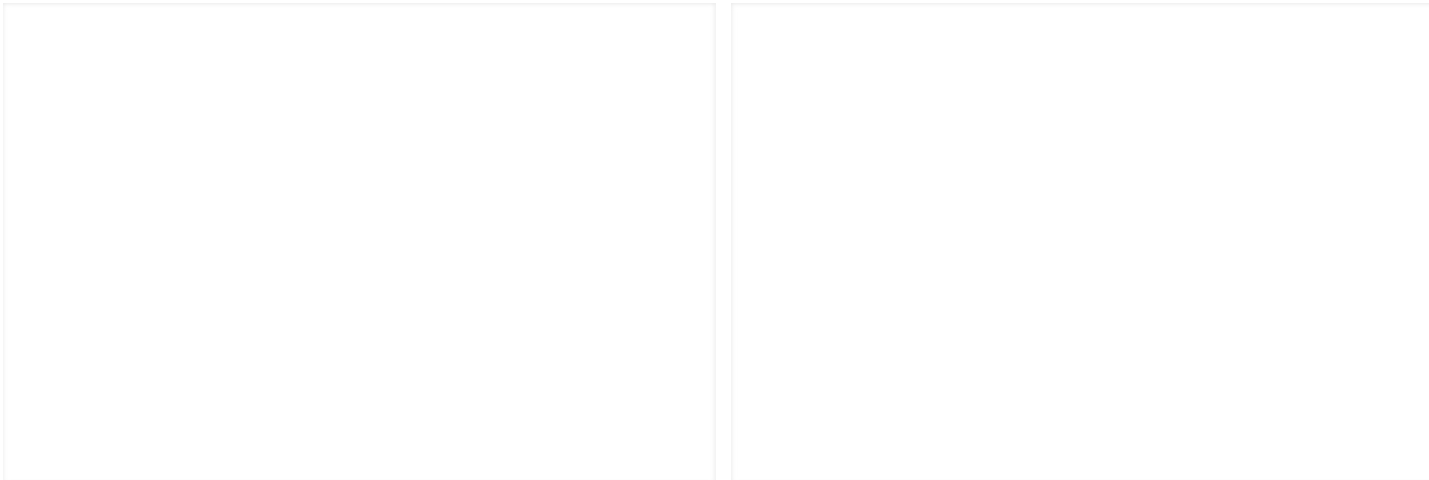
On the other hand, the second fall called "**Wang jok**" splitted from the main fall ie "Kabalega fall" due to flooding from the upper side of either side of the river, the water flows on top of a large underlying rocks of 123.98m width making a steam misty around the falling water zone.

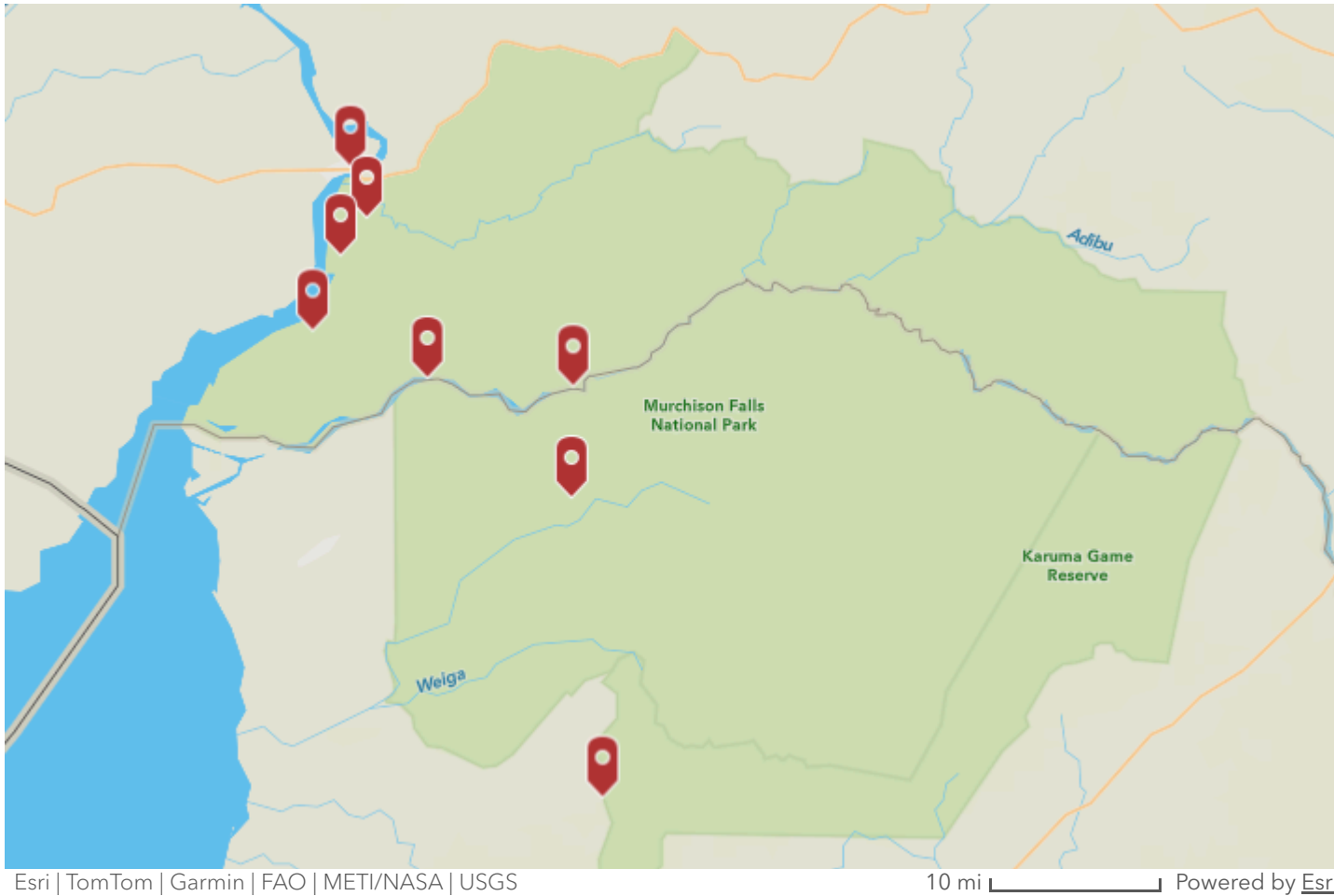


(Wang Jok)-Flood formed water fall from the main fall(2°16'44.69"N, 31°41'1.10"E)

Determination of the two falls using satellite imagery datasets

The two falls has got different strength of water flow and mode of formation, what determines the speed of water between the two falls is the vertical drop (height of the water fall) and the volume of water .From open source tool i.e. satellite imagery dataset ,at a distance length of cross section 724.96m there is Elevation gain over Elevation loss of (Elev G/ Elev L 70.4m, -73.6m) and a Maximum Slope of 35 % ,-42.7% meaning





Field Procedures



Day 1



Departure from Kampala



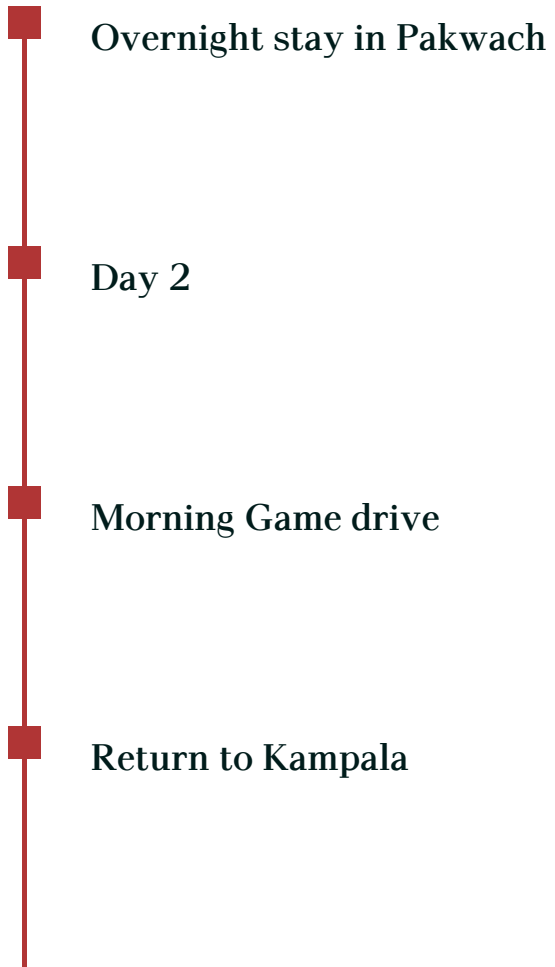
Stopovers in Masindi



Afternoon hike to Top of the
Falls



Arrival in Pakwach Town



Soil Classification Map of Murchison Falls NP



References

1. Uganda Tourism Satellite Account(2025).Statistical abstract trend.
2. Uganda Wildlife Authority (2023). Annual Conservation Status Report.
3. Uganda Wildlife Authority (UWA). (2024). Murchison Falls National Park: Visitor Information and Conservation Profile.
Kampala, Uganda
4. Kolb, D. A. (1984). Experiential Learning: Experience as the Source of Learning and Development.

5. OpenStreetMap contributors. (2025). Murchison Falls National Park boundary and features (Relation ID 2316689). OpenStreetMap. Retrieved September 30, 2025,
6. Ministry of Tourism, Wildlife & Antiquities (2023). Tourism Sector Review.
7. Protected Planet. (2025). Murchison Falls National Park (WDPA ID: 956). World Database on Protected Areas (WDPA). Retrieved September 30, 2025, from open street map.
8. Wildlife conservation challenges in Uganda's protected areas. African Journal of Ecology, 56(4), 801–809.
9. Plumptre, A. J., Davenport, T. R. B., Behangana, M., Kityo, R., Eilu, G., Ssegawa, P., ... & Ogwal, J. J. (2007). The biodiversity of the Albertine Rift. Biological Conservation, 134(2), 178–194.
10. Tumusiime, D. M., & Vedeld, P. (2015). Public protected areas, local people's livelihoods and conflicts: A case study from Murchison Falls National Park, Uganda. *Forest Policy and Economics*, 62, 61–70.

Field Data \$ Credit

This story map was developed from field observations collected during fieldwork in Murchison Falls National Park, Uganda.

<i>Fieldwork conducted</i>	<i>Welborn Kibwota-Researcher / Student Makerere University Geography Department</i>
<i>Field data collection</i>	<i>Date September 2025</i>
<i>Vegetation zonation based on field observations and supporting literature</i>	<i>Lock, J.M. Vegetation studies of Murchison Falls NP)</i>
<i>Photography</i>	<i>Welborn Kibwota \$ Okello Julius Peter</i>
<i>GIS base map and imagery</i>	<i>© Esri</i>